

Math 2551 Worksheet: Limits and Continuity

1. Let $f(x, y) = \left(\frac{1}{x} + \frac{1}{y}\right)^2$. Find $\lim_{(x,y) \rightarrow (2,-3)} f(x, y)$ or show it does not exist.
2. Let $f(x, y) = \frac{x - 2y}{x^3 - 8y^3}$. Find $\lim_{(x,y) \rightarrow (2,1)} f(x, y)$ or show it does not exist.
3. Let $f(x, y) = \frac{\sqrt{2x - y} - 2}{2x - y - 4}$. Find $\lim_{(x,y) \rightarrow (2,0)} f(x, y)$ or show it does not exist.
4. Let $f(x, y) = \frac{y^2}{x^2 + y^2}$. Find $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ or show it does not exist.
5. At what points (x, y) in the plane is $f(x, y) = \cos\left(\frac{1}{xy}\right)$ continuous?
6. At what points (x, y, z) is $h(x, y, z) = \frac{1}{1 - \ln(x^2 + y^2 + z^2)}$ continuous?